# 1. Clinical Question & Core Hypothesis

## Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

## Working Hypothesis

Primary hypothesis: In PD-1/PD-L1 inhibitor-treated bladder cancer, high TIL density predicts robust therapy response only when PD-L1 upregulation is adaptive (co-localized with T cells) and S100A9-driven myeloid inflammation is low. Conversely, when S100A9-positive myeloid inflammation is high, even TIL-high tumors exhibit reduced response due to myeloid-mediated T cell suppression, regardless of PD-L1 presence. The predictive value of TIL density is therefore context-dependent and enhanced when evaluated together with PD-L1 status and S100A9 levels, with further refinement by mutational load (bTMB) and other immune-context modifiers. Specifically, the following testable propositions arise: (1) In TIL-high tumors with low S100A9 and PD-L1 consistent with adaptive signaling (T cells present and PD-L1 upregulated in proximity to T cells), PD-1/PD-L1 blockade yields higher objective response rates (ORR) and longer duration of response compared with tumors with high S100A9. (2) In TIL-high but S100A9-high tumors, PD-1/PD-L1 blockade yields comparatively attenuated responses despite PD-L1 detection, due to myeloid-driven T cell suppression. (3) TIL density plus S100A9 and PD-L1 status provide superior predictive performance for therapy_response than any single marker, and this performance is further improved by incorporating bTMB and Immunoscore as secondary modifiers. (4) Spatial co-localization analyses showing PD-L1 expression on S100A9+ myeloid cells versus PD-L1 expression on T cells or tumor cells will differentially associate with response, supporting the adaptive vs non-adaptive PD-L1 dichotomy as a key determinant of ICI efficacy. (5) Immunoscore and methylation-based markers (SDC2, TFPI2) will independently modulate the TIL–PD-L1–S100A9 axis, refining risk stratification beyond the core 2-anchored model.

| Element | Value |
|---|---|
| Imaging feature | Tumor-Infiltrating Lymphocyte (TIL) density |
| Anchor gene(s) | S100A9, PD-L1 |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**S100A9:** S100A9→PD■L1 axis (Core Path 1) and the foundational TMB biomarker path (Core Path 2) because they provide a mechanistic chain linking tumor immunogenicity, myeloid-driven inflammation, checkpoint induction, and therapeutic vulnerability.

**PD-L1:** Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|

| | | |
|---|---|---|
| Secondary | bTMB | Supporting modifier from knowledge graph |
| Secondary | MSH6 | Supporting modifier from knowledge graph |
| Secondary | SDC2 | Supporting modifier from knowledge graph |
| Secondary | TFPI2 | Supporting modifier from knowledge graph |
| Secondary | Immunoscore | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

**Prediction 2:** At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSH6: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.